Air Quality Index (AQI) Classification Using Hybrid Multi-Input CNN

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***Abstract*—Air pollution has become one of the most critical environmental and public health challenges faced by modern urban societies. Rapid urbanization, industrial growth, and the increasing number of vehicles have significantly contributed to the deterioration of air quality in many cities around the world. Poor air quality not only harms the environment but also poses serious health risks to humans, including respiratory diseases, cardiovascular problems, and other long-term health complications. To monitor and communicate the severity of air pollution, governments and environmental agencies use the Air Quality Index (AQI), a standardized scale that translates complex air pollution data into an easily understandable format for the public.**

Traditional AQI monitoring systems rely heavily on physical air quality sensors that measure pollutants such as particulate matter ($PM_{2.5}$ and PM_{10}), nitrogen dioxide, sulfur dioxide, and carbon monoxide. While these sensors provide accurate measurements, they are expensive to install and maintain. Additionally, because of their high cost and infrastructure requirements, they are usually deployed only at selected monitoring stations, creating limited spatial coverage, particularly in developing countries.

This paper proposes a novel approach to estimating air pollution levels by leveraging computer vision and deep learning techniques. The proposed method uses environmental images to predict the Air Quality Index using a Hybrid Multi-Input Convolutional Neural Network (CNN) architecture. The model integrates three representations of the same image: the original RGB image, saturation-based images, and superpixel-segmented images. Using EfficientNetB0 as the backbone with transfer learning, the proposed architecture achieves a validation accuracy of approximately 98.67%, demonstrating the effectiveness of combining multiple image representations for scalable, cost-effective AQI estimation.

***Index Terms*—Air Quality Index, Convolutional Neural Network, EfficientNetB0, Transfer Learning, Superpixel Segmentation, Grad-CAM, Environmental Monitoring, Deep Learning.**

I. Introduction

Air pollution has emerged as one of the most pressing global environmental challenges of the modern era. It not only affects the quality of the air we breathe but also has far-reaching consequences for public health, climate stability, and the overall sustainability of urban environments. Over the past few decades, rapid industrialization, increased vehicular emissions, population growth, and uncontrolled urban expansion have significantly contributed to worsening air quality in many parts of the world. Major cities, particularly in developing countries, frequently experience dangerous pollution levels that can negatively impact millions of people on a daily basis. Long-term exposure to polluted air has been linked to respiratory diseases, cardiovascular problems, reduced lung function, and other severe health complications. Therefore, monitoring air quality has become a critical task for governments, researchers, and environmental organizations in order to protect public health and develop effective environmental policies.

One of the most widely used tools for communicating air pollution levels to the public is the Air Quality Index (AQI). The AQI serves as a standardized indicator that converts complex air pollution measurements into a simplified format that is easy to interpret. Instead of presenting raw pollutant concentration values, the AQI categorizes air quality into different levels such as *Good*, *Moderate*, *Unhealthy for Sensitive Groups*, *Unhealthy*, *Very Unhealthy*, and *Severe*. Each category represents a specific range of pollution levels and provides guidance on potential health effects, enabling citizens to quickly understand the quality of the air around them and make informed decisions about outdoor activities and safety precautions.

Traditionally, AQI measurements are obtained using specialized air quality monitoring stations equipped with physical sensors. These monitoring stations are capable of providing accurate and reliable data; however, installing and maintaining such infrastructure is often expensive and technically demanding. As a result, these monitoring systems are typically deployed only in selected locations within major cities, leading to significant gaps in spatial coverage that make it difficult to accurately capture variations in air quality across different neighborhoods or rural regions.

In recent years, advances in deep learning and computer vision have opened new possibilities for addressing these limitations. Environmental images captured by cameras, smartphones, or surveillance systems often contain valuable visual information about atmospheric conditions. Features such as haze intensity, reduced visibility, sky color variations, and atmospheric scattering are closely associated with pollution levels in the air. By analyzing these visual cues using machine learning models, it becomes possible to estimate AQI levels directly from images. This emerging approach has the potential to significantly enhance air quality monitoring capabilities by leveraging widely available camera networks and mobile devices to collect environmental data across larger geographical areas.

II. Problem Statement and Objectives

Despite significant advancements in air quality monitoring technologies, traditional sensor-based AQI measurement systems still face several practical limitations. Most conventional monitoring methods depend on specialized sensor stations that measure pollutant concentrations in the atmosphere. Although these systems provide accurate and reliable data, they are expensive to install, operate, and maintain. Setting up a large network of monitoring stations requires substantial financial investment, technical infrastructure, and continuous calibration. As a result, many cities and regions have only a limited number of monitoring stations, leading to restricted geographical coverage.

Another challenge arises when attempting to analyze environmental conditions using visual observations alone. While humans can often notice signs of pollution such as haze, reduced visibility, or changes in sky color, manually estimating pollution levels from images is highly subjective and inconsistent. Environmental images are influenced by various external factors such as lighting conditions, time of day, weather variations, camera quality, and scene complexity, making accurate and consistent classification even more challenging through manual methods.

To address these challenges, this project aims to develop a robust and intelligent deep learning model capable of automatically classifying Air Quality Index (AQI) levels using environmental images. The primary objectives of this study are:

1) Design and implement an effective hybrid multi-input CNN architecture capable of learning complex environmental features.
2) Enhance classification robustness by incorporating feature-enhanced image inputs that highlight important atmospheric characteristics.
3) Achieve high model performance, with strong validation accuracy during training and evaluation.
4) Incorporate explainable AI techniques, particularly Grad-CAM (Gradient-weighted Class Activation Mapping), to provide interpretable predictions.

Through these objectives, the project aims to contribute toward developing a more scalable, intelligent, and interpretable solution for air quality monitoring using modern deep learning technologies.

III. Literature Review

Over the past decade, the rapid advancement of deep learning and computer vision has encouraged researchers to explore new approaches for estimating air pollution using visual data. Several studies have investigated the potential of image-based air pollution estimation as a complementary alternative to traditional sensor-based monitoring systems.

Early research demonstrated that convolutional neural networks (CNNs) are highly effective for extracting meaningful visual features from environmental images [2]. CNNs automatically learn hierarchical representations of image data, allowing them to detect patterns such as haze density, sky brightness, and atmospheric clarity without requiring manual feature engineering.

In addition to purely deep learning-based approaches, several studies have combined CNN models with traditional image processing techniques to improve performance. Classical methods such as the Dark Channel Prior (DCP) and atmospheric scattering models have been widely used in image dehazing research [1]. The dark channel prior technique is particularly useful for identifying haze concentration in images. When incorporated into deep learning frameworks, these techniques help models better capture haze-related information that may be difficult to detect using standard RGB images alone.

More recent research has focused on hybrid CNN architectures that combine handcrafted features with automatically learned deep features [3]. Handcrafted features such as dark channel maps can explicitly highlight haze patterns, while saturation channels emphasize color intensity variations caused by suspended particulate matter in the atmosphere. Another important development is the use of multi-stream or multi-input CNN architectures, in which different representations of the same image are processed through separate neural network branches before being combined [5].

Furthermore, recent studies have explored lightweight deep learning architectures and autoencoder-based models for efficient pollution estimation [3]. Some researchers have also incorporated attention mechanisms into CNN architectures to help the model focus on the most informative regions of an image [5]. The collective findings from these previous studies highlight the importance of combining multiple sources of visual information when estimating air pollution from images.

Table I summarizes the key related works and their methodological contributions.

IV. Dataset Description

The model used in this study was trained using the Air Pollution Image Dataset, which consists of environmental images collected from various regions in India and Nepal. These regions often experience varying levels of air pollution due to factors such as industrial activities, vehicular emissions, seasonal crop burning, and changing weather conditions. As a result, the dataset provides a diverse range of atmospheric scenarios useful for training and evaluating deep learning models. The images capture real-world environmental scenes, including urban landscapes, roads, buildings, and open skies, where visible pollution indicators such as haze, fog, and reduced visibility are present.

To facilitate effective classification, the dataset categorizes images into six distinct AQI levels: *Good*, *Moderate*, *Unhealthy for Sensitive Groups*, *Unhealthy*, *Very Unhealthy*, and *Severe*. This multi-class categorization enables the model to learn nuanced differences between various pollution levels

TABLE I
SUMMARY OF RELATED LITERATURE

Paper Title	**Authors**	**Methodology**	**Key Findings**
Image-based Air Pollution Estimation Using Hybrid CNN (2019)	Jian Ma, Kun Li, Yahong Han	Hybrid Deep CNN + Dark Channel Features + Feature Fusion	Dark channel captures haze information. Multi-stream fusion improves robustness.
Image-based Air Quality Analysis Using Deep CNN (2020)	Yilmar Chakma, Ben Vizena, Tingting Cao, Jerry Lin, Jing Zhang	Deep CNN + Transfer Learning	CNN learns features automatically. Transfer learning improves accuracy. No handcrafted features needed.
Image-Based PM Pollution Analysis Using Lightweight Autoencoder (2021)	Xulong Zhang, Deepak Shrestha, Afsana Khan, Jing Zhang	Lightweight Autoencoder + Saturation Channel + Superpixels	Saturation is a useful pollution indicator. Superpixels preserve structure. Efficient lightweight design.
Multispectral Satellite Image Dehazing & Pollution Estimation (2018–2019)	Sai Avinash G, Shaistha M, Nishen Ganegoda, John Basha M., Naga Kushal Ageeru	Dark Channel Prior + Atmospheric Scattering Model	Haze & pollution strongly correlated. Transmission maps improve estimation.
Two-Stream Non-Uniform Concentration Reasoning Network (2021–2022)	Huilin Chen, Wenming Yang, Qingmin Liao	Two-Stream CNN + Attention Mechanism	Multi-stream improves feature capture. Attention enhances performance.

rather than simply distinguishing between clean and polluted air.

A. Preprocessing

Before training the neural network, several preprocessing steps were applied to ensure consistency and compatibility with the deep learning architecture. Each image in the dataset was resized to a standardized resolution of 224×224 pixels. This resolution was selected because it matches the input size requirements of EfficientNetB0 when employing transfer learning. After resizing, normalization was applied to scale pixel values to a consistent range, which helps stabilize the training process and improves convergence during optimization.

B. Data Augmentation

Data augmentation techniques were incorporated to enhance dataset diversity and improve the model's ability to generalize to unseen images. In this project, several augmentation methods were used, including image rotation, horizontal flipping, and scaling transformations. These operations simulate variations that might naturally occur in real-world images, such as changes in camera orientation, perspective, or scene composition.

C. Class Imbalance Handling

One of the key challenges encountered was class imbalance within the dataset. Certain AQI categories contained significantly more images than others, which could potentially bias the model toward predicting the majority classes more frequently. To address this issue, a class weighting strategy was implemented during the training phase. By assigning higher weights to minority classes and lower weights to majority classes, the training process encourages the model to pay greater attention to less represented categories.

V. METHODOLOGY

The proposed system is built around a Hybrid Multi-Input CNN architecture specifically designed to capture complementary information from multiple representations of environmental images.

A. Input Representations

The system generates three distinct image representations to provide the network with complementary atmospheric information.

1) RGB Image: The original RGB image preserves the natural visual appearance of the environment, serving as the primary input stream and capturing the overall atmospheric scene.

2) Saturation Image: The saturation representation highlights variations in color intensity within the image. Atmospheric pollution often affects the color characteristics of the sky and surrounding environment due to the presence of particulate matter that scatters light. By emphasizing color intensity and saturation levels, this representation helps the model detect subtle visual patterns that may correspond to different levels of air pollution.

3) Superpixel Image: The superpixel segmentation divides the original image into groups of pixels that share similar visual characteristics, such as color or texture. Superpixel images allow the neural network to focus on broader atmospheric patterns such as haze distribution and visibility degradation across different parts of the image, while reducing noise and unnecessary detail.

B. Network Architecture

To further enhance the model's performance, transfer learning is employed using EfficientNetB0 as the backbone architecture [6]. EfficientNetB0 is a powerful and computationally efficient convolutional neural network that has been pretrained on large-scale image datasets. Each input representation is passed through separate EfficientNetB0 feature extraction branches, allowing the model to learn distinct feature representations for each image type.

After feature extraction, the outputs from the three input streams are combined through feature concatenation, creating a unified feature representation that integrates information from all input sources. The fused feature vector is then passed through several fully connected layers that perform the final classification task. To improve training stability and prevent overfitting, batch normalization and dropout regularization are applied within these layers. Finally, the output layer predicts the corresponding AQI category using a softmax activation function.

The overall architecture can be summarized as:

$$\mathbf{f}_{fused} = \text{Concat}(f_{\text{RGB}},\, f_{\text{sat}},\, f_{\text{sp}}) \tag{1}$$

$$\hat{y} = \text{softmax}(W \cdot \text{Dropout}(\text{BN}(\mathbf{f}_{fused})) + b) \tag{2}$$

where f_{RGB}, f_{sat}, and f_{sp} are the feature vectors extracted from the RGB, saturation, and superpixel branches respectively, W is the weight matrix of the classification head, and b is the bias term.

VI. Model Improvements

During the early stages of experimentation, the initial version of the model achieved a validation accuracy of approximately 52%. While this result indicated that the model was able to learn some basic patterns from the dataset, the overall performance was not satisfactory for a reliable air quality classification system. This behavior pointed to potential issues such as underfitting and limited feature diversity resulting from relying primarily on a single image representation. Several enhancements were introduced to overcome these limitations.

A. Transfer Learning with EfficientNetB0

The most significant improvement was the incorporation of transfer learning using the EfficientNetB0 architecture. Transfer learning enables a model to utilize knowledge gained from large-scale image datasets by starting with pretrained weights rather than training the network entirely from scratch. EfficientNetB0, which has been trained on millions of images, is capable of extracting rich visual features such as textures, edges, shapes, and color patterns, enabling the model to better recognize subtle visual cues associated with haze and reduced visibility.

B. Hybrid Multi-Input Architecture

Another important improvement involved redesigning the architecture to adopt a hybrid multi-input framework. Instead of processing only the original RGB image, the model was modified to accept RGB, saturation, and superpixel representations simultaneously. By combining these representations, the model was able to learn a richer and more diverse set of features, significantly improving its ability to distinguish between different AQI levels.

C. Regularization and Training Stability

Dropout regularization was introduced in the fully connected layers to prevent the network from becoming overly dependent on specific neurons during training. Batch normalization was also applied to stabilize the training process by maintaining consistent activation distributions across network layers.

D. Class Weighting and Fine-Tuning

To address class imbalance, class weighting was introduced during the training process. Additionally, fine-tuning of the pretrained EfficientNetB0 layers was performed using a small learning rate, allowing the pretrained model to adapt its feature representations to the specific characteristics of the air pollution image dataset without disrupting the valuable pretrained knowledge.

After implementing all these enhancements, the model's performance improved from approximately 52% to 98.67% validation accuracy, as summarized in Table II.

TABLE II
MODEL PERFORMANCE ACROSS CONFIGURATIONS

Model Configuration	**Validation Accuracy**
Initial Baseline (single RGB input)	$\approx 52\%$
With Transfer Learning (EfficientNetB0)	$\approx 85\%$
Hybrid Multi-Input + Fine-Tuning	$\approx 98.67\%$

VII. Results and Analysis

The experimental results obtained from the proposed hybrid CNN model demonstrate that the system performs highly effective AQI classification using environmental images. The evaluation of the model was conducted using several performance indicators, including training accuracy, validation accuracy, validation loss, F1 score, and confusion matrix analysis.

A. Accuracy and Loss Curves

One of the key observations during training was the behavior of the accuracy curves over multiple training epochs. Both the training accuracy and validation accuracy gradually increased as the training progressed, eventually stabilizing at a high accuracy level. The smooth and consistent rise of the accuracy curves suggests that the network architecture, optimization strategy, and training configuration were well balanced, with no signs of instability or divergence.

Throughout the training process, the validation loss remained consistently low and gradually decreased as the model improved its predictions. A low validation loss indicates that the model is making accurate predictions on unseen validation data and is not simply memorizing the training dataset, confirming good generalization capability.

B. F1 Score and Confusion Matrix

To further evaluate classification performance across the different AQI categories, additional evaluation metrics such as precision, recall, and F1 score were analyzed. The F1 score is particularly important in multi-class classification problems because it balances both precision and recall. The results showed strong and consistent F1 scores across most AQI categories, indicating that the model performs well not only for the majority classes but also for categories with fewer training samples.

The confusion matrix revealed that most images were correctly classified into their respective categories. Only a small number of misclassifications were observed, typically between neighboring AQI categories that share similar visual characteristics, such as *Moderate* and *Unhealthy for Sensitive Groups*. Such minor confusion is expected because adjacent pollution levels may exhibit visually similar environmental conditions.

C. Discussion

Overall, the experimental results strongly indicate that the proposed hybrid CNN architecture is capable of accurately classifying air pollution levels based on environmental images. The combination of stable training convergence, low validation loss, strong F1 scores, and minimal misclassification in the confusion matrix highlights the reliability and robustness of the model. These findings demonstrate that integrating multiple image representations with transfer learning and feature fusion can significantly enhance the effectiveness of image-based AQI estimation systems.

VIII. Explainability Using Grad-CAM

In addition to achieving high classification accuracy, it is also important to understand how the deep learning model makes its predictions. Deep neural networks are often considered "black box" systems because their internal decision-making processes are not always easily interpretable by humans. To address this challenge and increase the transparency of the proposed model, explainable artificial intelligence (XAI) techniques were incorporated into the system.

A. Grad-CAM Methodology

One of the most widely used methods for interpreting convolutional neural networks is Gradient-weighted Class Activation Mapping, commonly known as Grad-CAM [7]. Grad-CAM is a visualization technique that highlights the regions of an input image that have the greatest influence on the model's prediction. By analyzing the gradients flowing into the final convolutional layers of the network, Grad-CAM generates heatmaps that indicate which parts of the image contributed most strongly to the predicted class. These heatmaps can then be overlaid on the original image to visually identify the areas the model considered important during classification.

The Grad-CAM heatmap $L^c_{\text{Grad-CAM}}$ for class c is computed as:

$$L^c_{\text{Grad-CAM}} = \text{ReLU}\left(\sum_k \alpha^c_k A^k\right) \tag{3}$$

where A^k denotes the activation map of the k-th feature map in the final convolutional layer, and α^c_k is the importance weight for class c obtained by global average pooling of the gradients:

$$\alpha^c_k = \frac{1}{Z}\sum_i \sum_j \frac{\partial y^c}{\partial A^k_{ij}} \tag{4}$$

B. Grad-CAM Observations

In this project, Grad-CAM was applied to several validation images after the model completed the training process. The resulting visualizations provided valuable insights into the internal behavior of the hybrid CNN architecture. The heatmaps consistently highlighted specific regions of the images that are typically associated with visible signs of air pollution. In many cases, the model focused on areas of the sky where haze concentration was most apparent. Regions with dense atmospheric haze, reduced visibility, or scattered light patterns were frequently emphasized in the Grad-CAM visualizations.

Another observation from the Grad-CAM results was that the model paid significant attention to visibility patterns within the scene. Areas where distant objects such as buildings, mountains, or skyline features appeared blurred or partially obscured were often highlighted by the activation maps, consistent with the visual effects of suspended particulate matter in the atmosphere. Additionally, the Grad-CAM heatmaps often emphasized regions with noticeable contrast degradation and color distortion, confirming that the model focuses on pollution-relevant visual features.

Overall, the Grad-CAM visualizations confirmed that the neural network was not making arbitrary predictions but was instead relying on relevant environmental features present in the images. This transparency not only increases trust in the system but also makes it easier for researchers and users to understand and interpret the behavior of the deep learning model in practical air quality monitoring applications.

IX. Future Scope

A. Completed Milestones

Since the initial submission of this progress report, several components that were previously identified as future work have been successfully implemented and deployed.

1) Mobile-Friendly Web Application: A fully functional web-based interface has been developed and deployed, enabling users to upload environmental images and receive real-time AQI predictions in simplified categorical classes — *Healthy*, *Moderate*, and *Unhealthy* — without requiring any local setup. The application is cross-device accessible and optimized for both desktop and mobile browsers.

2) Cloud Deployment and Backend Infrastructure: The backend has been deployed on cloud infrastructure, ensuring scalability and real-time processing. The complete backend source code is publicly available at:

https://github.com/ASR222/AQI_project

The repository includes the trained model weights, FastAPI-based inference endpoints, and deployment configuration for reproducibility.

B. Current Future Scope

1) Video-Based AQI Classification: The next major technical milestone is extending the current image-based classification pipeline to support continuous video streams. Rather than classifying individual static frames, the proposed system will perform temporal aggregation across consecutive frames to produce smoother and more reliable AQI estimates in real time. This will involve integrating lightweight temporal models or sliding-window inference over live camera feeds, making the system suitable for deployment on fixed surveillance infrastructure, traffic cameras, and drone footage.

2) Government and Institutional Tie-Ups: We are actively pursuing collaborations with relevant government bodies and environmental agencies to integrate our image-based AQI estimation system as a supplementary tool within existing air quality monitoring networks. Such partnerships would enable access to geographically distributed camera infrastructure, ground-truth sensor data for model validation, and pathways toward official recognition of image-based monitoring as a credible supplementary methodology. The goal is to bridge the gap between academic research and actionable environmental policy through institutional adoption.

X. Conclusion

This paper demonstrates the strong potential of deep learning techniques for performing air quality classification using environmental images. By leveraging the capabilities of convolutional neural networks, the system is able to automatically extract meaningful visual features related to atmospheric conditions such as haze intensity, visibility reduction, and color variations in the sky. Unlike traditional approaches that rely heavily on physical air quality sensors, this image-based method offers a more flexible and scalable way to estimate air pollution levels using widely available visual data.

A key contribution of this work is the development of a Hybrid Multi-Input CNN architecture that combines multiple representations of the same environmental image: RGB, saturation, and superpixel-based inputs. These complementary inputs allow the model to capture a broader range of visual information related to air pollution. The use of transfer learning with EfficientNetB0 further enhances the model's ability to recognize subtle environmental patterns associated with different pollution levels, while Grad-CAM visualization provides interpretable insights into the model's decision-making process.

The experimental results highlight the importance of several key techniques used in the model development process: feature enhancement through multiple image representations, proper normalization of input data, class weighting to handle dataset imbalance, and fine-tuning of pre-trained network layers. Together, these strategies contribute significantly to the model's performance, achieving a validation accuracy exceeding 98.67% across multiple AQI categories.

Overall, the system shows strong potential as a scalable and cost-effective alternative to conventional sensor-based air quality monitoring methods. By utilizing environmental images captured from cameras or mobile devices, the approach could enable broader geographic coverage and more accessible air quality analysis. This work therefore contributes to the development of intelligent, data-driven solutions for environmental monitoring and highlights the promising role of deep learning in addressing real-world air pollution challenges.

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